Using AI to Explore Research Ideas

- I use AI to explain unfamiliar papers and compare existing methods.
- I describe a research problem and ask AI to suggest possible solutions.
- I challenge its suggestions: Is the idea new? Is it practical? What are its weaknesses?
- I refine the problem through several rounds of discussion.

Turning Ideas into Experiments

- Once I select an idea, I ask AI to prepare a detailed implementation plan.
- I use a coding assistant, such as Codex, to help turn the plan into experimental code.
- I run experiments and compare the method with existing approaches, including simple baselines.
- I share the results with AI to discuss findings and possible improvements.

My AI Research Setup

Three computers. Two locations. One research workflow.



- **Code and experiments run on the Ubuntu desktop in the lab.**

Improving Communication and Verification

- I use AI to organize research reports, prepare slides, and improve my academic English writing.
- I ask it to clarify explanations in the reasoning.
- I discuss important decisions with my supervisor.